“Intelligent Water Distribution Monitoring and Leakage Detection Platform”

CO1 Assessment Tool 2

Submitted in the partial fulfilment for the Course of

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DECLARATION

I, **Lenin K 192321050,** of the **Department of Information Technology**, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **"Collaborative Story/ Screenplay Writing Platform with Version History"** is the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place:

Chennai

Date:

Signature of the Students

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1. Project Vision & Stakeholder Identification

1.1 Project Vision

The Intelligent Water Distribution Monitoring and Leakage Detection Platform is a cloud-based, IoT-enabled software system that gives water utilities continuous, real-time visibility into the health of their distribution networks. The platform ingests live flow, pressure, and acoustic sensor data from across the pipeline network, automatically detects and localizes probable leaks using anomaly-detection analytics, and predicts high-risk pipe segments before they fail. Integrated GIS mapping, instant alerting, and mobile work-order management let control-room operators and field crews move from detection to repair with minimal delay. The vision is to become the trusted operational backbone that helps utilities reduce non-revenue water (NRW) loss, extend asset life, and deliver a more reliable water supply to the communities they serve.

1.2 Project Objectives

- Enable continuous, real-time ingestion and monitoring of flow, pressure, and acoustic sensor data across the entire distribution network.
- Automatically detect and localize probable leaks using anomaly-detection analytics on live sensor streams.
- Predict high-risk pipeline segments before failure using historical data and machine learning-based risk scoring.
- Convert detected leaks into actionable, trackable work orders for field repair crews.
- Provide GIS-based visualization of the pipeline network integrated with live sensor and leak data.
- Support regulatory and management reporting on water loss, leak response times, and asset condition.

1.3 Key Stakeholders

Stakeholder	Role / Interest in the Project
Control Room Operators (Primary Users)	Monitor the real-time dashboard, validate leak alerts, and dispatch field crews.
Field Technicians / Repair Crews	Receive assigned work orders, locate and repair leaks, and update job status on-site.
Utility / Asset Managers	Review risk scores, prioritize maintenance budgets, and track NRW performance over time.
GIS / Network Engineers	Maintain the digital twin of the pipeline network and keep spatial data synchronized.
Data Scientists / ML Engineers	Build and tune the leak-detection and pipe-failure-risk models.
Municipal / Regulatory Authorities	Require periodic water-loss and compliance reporting from the utility.
Consumers / Customers	Indirectly benefit from fewer outages and can report suspected leaks or low pressure.

Stakeholder	Role / Interest in the Project
IoT Sensor Vendors / Hardware Integrators	Supply and maintain the flow, pressure, and acoustic sensor hardware and data feeds.
Platform Administrator	Manages utility accounts, user roles, zone configuration, and platform security.
Product Owner	Owns the product backlog, defines priorities, and represents stakeholder interests.
Scrum Master	Facilitates Agile ceremonies and removes impediments for the development team.
Development Team (Frontend / Backend / DevOps)	Designs, builds, tests, and deploys the platform's features.
QA / Test Engineers	Validate the accuracy of sensor ingestion, leak detection, and alerting workflows.

2. Epics and User Stories

The product requirements have been decomposed into seven Epics, each representing a major functional area of the platform. Every Epic is further broken down into well-structured User Stories following the standard Agile format: “As a <user>, I want <feature>, so that <benefit>.”

EPIC-01: User Accounts, Utility & Zone Management

Epic Goal: Allow utility staff to register, set up their organization, and define distribution zones/DMAs.

US-01: As a new utility admin, I want to create an account and register my utility organization, so that I can start configuring my water distribution network on the platform. *(Estimated: 3 pts)*

US-02: As a utility admin, I want to define District Metered Areas (DMAs) and pipeline zones on the network, so that monitoring and leak detection can be organized by manageable sub-regions. *(Estimated: 5 pts)*

US-03: As a utility admin, I want to invite team members and assign roles (Operator, Field Technician, Manager, Analyst), so that the right people have appropriate access to monitoring and repair tools. *(Estimated: 5 pts)*

US-04: As a utility admin, I want to remove or change the role of a team member at any time, so that I retain control over system access as staff changes. *(Estimated: 3 pts)*

EPIC-02: Real-Time Sensor Data Ingestion & Monitoring Dashboard

Epic Goal: Ingest live data from IoT flow, pressure, and acoustic sensors and present it on a real-time monitoring dashboard.

US-05: As the system, I want to ingest streaming flow, pressure, and acoustic sensor data from IoT devices installed across the network, so that the platform has continuous, up-to-date visibility into pipeline conditions. *(Estimated: 8 pts)*

US-06: As a control room operator, I want to view a real-time dashboard of flow and pressure readings across all DMAs, so that I can monitor overall network health at a glance. *(Estimated: 8 pts)*

US-07: As an operator, I want to drill down from a zone-level view into individual sensor/meter readings, so that I can investigate a specific area in more detail. *(Estimated: 5 pts)*

US-08: As an operator, I want the system to automatically flag sensors that go offline or report faulty readings, so that data gaps are not mistaken for real anomalies. *(Estimated: 3 pts)*

EPIC-03: Leak Detection & Localization Engine

Epic Goal: Automatically detect and localize probable leaks using anomaly detection on sensor data streams.

US-09: As the system, I want to continuously analyze flow and pressure data for statistical anomalies (e.g., minimum night flow deviations), so that potential leaks are flagged automatically without manual monitoring. *(Estimated: 13 pts)*

US-10: As an operator, I want to see a probable leak location highlighted on the network map with a confidence score, so that I can quickly assess where a suspected leak is occurring. *(Estimated: 8 pts)*

US-11: As an operator, I want to view the sensor readings and anomaly trend that triggered a leak alert, so that I can validate whether the alert is genuine before dispatching a crew. *(Estimated: 5 pts)*

US-12: As an operator, I want to manually mark a flagged anomaly as a confirmed leak, false positive, or scheduled maintenance, so that the detection model improves over time and history stays accurate. *(Estimated: 5 pts)*

EPIC-04: Predictive Maintenance & Analytics

Epic Goal: Use historical data and machine learning to predict pipe failures before they occur and support long-term asset planning.

US-13: As a utility manager, I want the system to calculate a pipe-failure risk score for each network segment based on age, material, pressure history, and past incidents, so that I can prioritize proactive maintenance and replacement. *(Estimated: 13 pts)*

US-14: As a utility manager, I want to view a ranked list of high-risk pipeline segments, so that I can plan preventive maintenance budgets and schedules. *(Estimated: 5 pts)*

US-15: As an analyst, I want to view historical trends in non-revenue water (NRW) loss by zone, so that I can measure the impact of leak-reduction efforts over time. *(Estimated: 5 pts)*

EPIC-05: Alerts, Work Orders & Field Crew Management

Epic Goal: Turn detected leaks into actionable work orders and keep field crews and managers informed in real time.

US-16: As an operator, I want to receive an instant alert (push/SMS/email) when a high-confidence leak is detected, so that I can respond and dispatch a crew without delay. *(Estimated: 5 pts)*

US-17: As an operator, I want to create and assign a work order to a field technician directly from a leak alert, so that repair actions are tracked from detection to resolution. *(Estimated: 8 pts)*

US-18: As a field technician, I want to view my assigned work orders with the leak location and details on a mobile-friendly interface, so that I can navigate to the site and begin repairs efficiently. *(Estimated: 5 pts)*

US-19: As a field technician, I want to update a work order's status (En Route, In Progress, Resolved) and attach repair notes/photos, so that the utility has an accurate record of the repair. *(Estimated: 5 pts)*

EPIC-06: GIS Mapping & Network Visualization

Epic Goal: Provide a geospatial view of the entire pipeline network integrated with sensor and leak data.

US-20: As an operator, I want to view the full pipeline network overlaid on a GIS map with pipe age, material, and diameter, so that I have spatial context for every monitoring or leak event. *(Estimated: 8 pts)*

US-21: As a GIS engineer, I want to import and update pipeline network geometry from shapefiles/GeoJSON, so that the platform's digital twin stays synchronized with as-built infrastructure changes. *(Estimated: 8 pts)*

EPIC-07: Reporting, Compliance & Data Export

Epic Goal: Provide reporting and export capability for regulatory compliance and management review.

US-22: As a utility manager, I want to generate and export monthly water-loss and leak-response reports (PDF/CSV), so that I can meet regulatory reporting requirements and share performance with stakeholders. *(Estimated: 5 pts)*

3. Acceptance Criteria (Given–When–Then)

Clear, testable acceptance criteria have been written for the highest-priority user stories selected for the first sprint (and key stories from later sprints), using the Given–When–Then format to ensure each requirement is unambiguous and verifiable by QA.

US-02: Define a District Metered Area (DMA)

Given a logged-in utility admin on the network setup screen,

When the admin draws a zone boundary on the map and saves it as a new DMA,

Then the system creates the DMA, assigns it a unique identifier, and associates all sensors within its boundary to that zone.

Given a DMA already exists with overlapping boundaries,

When the admin attempts to save the new zone,

Then the system warns of the overlap and requires the admin to resolve it before saving.

US-05: Ingest streaming sensor data

Given an IoT flow/pressure sensor is registered and online,

When the sensor transmits a new reading,

Then the platform ingests the reading within 5 seconds and stores it with sensor ID, timestamp, and zone tag.

Given a sensor stops transmitting for more than 2 minutes,

When the ingestion service checks its heartbeat, Then the sensor is marked ‘Offline’ and excluded from live anomaly analysis until it reconnects.

US-06: Real-time monitoring dashboard

Given a control room operator opens the dashboard,

When the dashboard loads,

Then the operator sees current flow and pressure values for every DMA, refreshed at least every 10 seconds, with zones color-coded by status (Normal, Warning, Alert).

Given a DMA's readings cross a configured anomaly threshold,

When the dashboard updates,

Then that DMA's tile changes color and moves to the top of the priority list.

US-09: Automatic leak anomaly detection

Given a DMA has at least 14 days of baseline flow and pressure history,

When the current minimum night flow or pressure deviates beyond the statistically expected range for that zone,

Then the system raises an anomaly flag with a calculated confidence score and timestamps the deviation.

Given the deviation is within normal seasonal or demand-driven variation,

When the anomaly-detection engine evaluates the reading,

Then no false alert is raised, and the reading is logged for model retraining.

US-11: Validate a leak alert

Given an operator opens a raised leak alert,

When the alert detail view loads,

Then the operator sees the flow/pressure trend chart, the anomaly confidence score, and the affected DMA highlighted on the map.

Given the trend chart shows the same anomaly pattern for more than 3 consecutive readings,

When the operator reviews the data,

Then the confidence score is displayed as 'High' to support a faster go/no-go decision.

US-16: Instant alert on high-confidence leak

Given the anomaly-detection engine raises a leak flag with a confidence score above the configured high-confidence threshold,

When the flag is confirmed,

Then the system sends a push notification, SMS, and email to all on-duty operators for the affected zone within 30 seconds.

Given no operator acknowledges the alert within 10 minutes,

When the escalation timer expires,

Then the alert is automatically escalated to the utility manager.

US-20: GIS network visualization

Given an operator opens the network map view,

When the map loads,

Then the operator sees all pipeline segments color-coded by material or age, with sensors and any active leak alerts plotted at their correct geolocation.

Given the operator clicks a pipeline segment,

When the segment is selected,

Then a side panel shows its age, material, diameter, and current risk score.

US-22: Export a monthly water-loss report

Given a utility manager opens the Reports section,

When the manager selects a date range and clicks ‘Export’,

Then the system generates a PDF and CSV report summarizing NRW volume, leak counts, average response time, and repair status by zone.

Given the export completes,

When the operator downloads the generated file,

Then the file is available for download and is also archived under the utility's compliance-reporting history.

4. Product Backlog Prioritization

The full product backlog was prioritized using a combination of MoSCoW categorization and a 1–10 Business Value score, taking into account customer needs, technical dependencies, and the logical build order of the platform (e.g., utility/zone setup and sensor ingestion must exist before dashboards, leak detection, or alerting are usable).

ID	Epic	User Story (short)	Value (1-10)	MoSCoW	Dependency
US-01	EPIC-01	create an account and register my utility organization	9	Must	None
US-02	EPIC-01	define District Metered Areas (DMAs) and pipeline zones	9	Must	US-01
US-05	EPIC-02	ingest streaming flow, pressure, and acoustic sensor data	10	Must	US-02
US-06	EPIC-02	view a real-time dashboard of flow and pressure readings	9	Must	US-05
US-03	EPIC-01	invite team members and assign roles	8	Must	US-02
US-08	EPIC-02	auto-flag sensors that go offline or report faulty data	7	Must	US-05
US-09	EPIC-03	detect flow/pressure anomalies automatically	10	Must	US-06
US-16	EPIC-05	receive an instant alert on a high-confidence leak	8	Should	US-09
US-10	EPIC-03	see a probable leak location on the map with confidence score	7	Should	US-09
US-17	EPIC-05	create and assign a work order from a leak alert	7	Should	US-16
US-11	EPIC-03	view the readings/trend that triggered a leak alert	7	Should	US-09
US-07	EPIC-02	drill down into individual sensor/meter readings	6	Should	US-06
US-18	EPIC-05	view assigned work orders on a mobile interface	6	Should	US-17
US-20	EPIC-06	view the pipeline network on a GIS map	6	Should	US-02
US-19	EPIC-05	update work order status and attach repair notes/ photos	5	Should	US-17
US-04	EPIC-01	remove or change the role of a team member	5	Should	US-03
US-12	EPIC-03	mark an anomaly as confirmed leak / false positive	5	Should	US-11
US-13	EPIC-04	calculate a pipe-failure risk score per segment	6	Could	US-09
US-21	EPIC-06	import/update pipeline geometry from shapefiles/ GeoJSON	5	Could	US-20
US-14	EPIC-04	view a ranked list of high-risk pipeline segments	4	Could	US-13

ID	Epic	User Story (short)	Value (1-10)	MoSCoW	Dependency
US-15	EPIC-04	view historical NRW loss trends by zone	4	Could	US-09
US-22	EPIC-07	export monthly water-loss and leak-response reports	4	Could	US-15

5. Sprint Backlog & Task Breakdown

Sprint 1 (2-week sprint) focuses on establishing the platform's foundation: utility onboarding, DMA/zone setup, team access control, and the core real-time sensor-ingestion and monitoring dashboard. These stories were selected because they are all rated 'Must' items that unblock nearly every other Epic in subsequent sprints — leak detection, alerting, and analytics all depend on live, reliable sensor data flowing into a working dashboard first.

5.1 Selected Sprint 1 Stories

ID	User Story	Story Points
US-01	As a new utility admin, I want to create an account and register my utility organization…	3
US-02	As a utility admin, I want to define District Metered Areas (DMAs) and pipeline zones…	5
US-03	As a utility admin, I want to invite team members and assign roles…	5
US-04	As a utility admin, I want to remove or change the role of a team member…	3
US-05	As the system, I want to ingest streaming flow, pressure, and acoustic sensor data…	8
US-06	As a control room operator, I want to view a real-time dashboard of flow and pressure readings…	8
US-07	As an operator, I want to drill down into individual sensor/meter readings…	5
US-08	As an operator, I want the system to auto-flag sensors that go offline or report faulty readings…	3

5.2 Task Breakdown per Story

US-01: Create an account and register my utility organization

- Design registration & login UI (Frontend)
- Implement email/password + OAuth (Google) sign-up API (Backend)
- Build utility organization setup page (name, service area, contact details)
- Write unit & integration tests for the auth flow

US-02: Define District Metered Areas (DMAs) and pipeline zones

- Design the map-based DMA-drawing tool (Frontend)
- Define data schema for zones, boundaries, and sensor-to-zone association
- Implement backend endpoint to save and validate zone boundaries (overlap checks)
- Load zone boundaries onto the network map view

US-03: Invite team members and assign roles

- Design collaborator invite modal with role dropdown (Operator, Technician, Manager, Analyst)
- Implement email invitation service (send + accept link)
- Implement role-based access control (RBAC) middleware
- Test invite, accept, and role-enforcement scenarios

US-04: Remove or change the role of a team member

- Add role-edit and remove-member controls to the team management screen
- Implement backend endpoint to update or revoke access
- Ensure active sessions are invalidated on role change/removal
- Test permission changes take effect immediately

US-05: Ingest streaming flow, pressure, and acoustic sensor data

- Build the sensor-data ingestion API/message broker (MQTT/Kafka-based)
- Implement schema validation and time-series storage for sensor readings
- Implement sensor registration and zone-tagging logic
- Load/stress test ingestion with simulated high-frequency sensor traffic

US-06: Real-time monitoring dashboard

- Design the zone-level dashboard UI with live status tiles
- Implement WebSocket/streaming layer to push live readings to the frontend
- Implement zone status color-coding logic (Normal/Warning/Alert)
- Cross-browser and performance QA of the live dashboard

US-07: Drill down into individual sensor/meter readings

- Design the sensor-detail drill-down panel
- Implement backend endpoint to fetch historical readings for a single sensor
- Add time-range selector and chart visualization for sensor trends
- Test navigation between zone view and sensor detail view

US-08: Auto-flag offline or faulty sensors

- Implement heartbeat/last-seen tracking per sensor
- Implement rule-based fault detection (stuck values, out-of-range readings)
- Add ‘Offline/Faulty’ status badge to the dashboard and sensor list
- Test that flagged sensors are excluded from anomaly analysis

6. Story Point Estimation Using Story Points

Story points were assigned using Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13, 21), where the team estimated relative effort, complexity, and uncertainty rather than absolute time. The whole team (Product Owner, Scrum Master, Developers, QA, and a Data/ML Engineer) independently proposed estimates; divergent votes were discussed and re-voted until consensus was reached, following standard Planning Poker practice.

6.1 Estimation Scale Reference

Points	Relative Complexity Meaning
1	Trivial change, no unknowns (e.g., copy/label update).
2	Small, well-understood task with minimal integration.
3	Small feature with minor UI/backend work and clear requirements.
5	Medium feature involving both frontend and backend, some design decisions.
8	Complex feature with real-time data, multiple edge cases, or new architecture.
13	Very complex feature with significant unknowns, cross-cutting impact, or algorithmic difficulty.

6.2 Story Point Justification (Sprint 1 Stories)

US-01: 3 pts — standard auth flow using well-known libraries; low uncertainty, minimal integration risk.

US-02: 5 pts — requires an interactive map-drawing tool and zone/sensor association logic; moderate design decisions.

US-03: 5 pts — needs email invitation service plus role-based access control; moderate backend complexity.

US-04: 3 pts — builds directly on existing RBAC; well-understood update/revoke pattern.

US-05: 8 pts — real-time, high-frequency ingestion pipeline (MQTT/Kafka) with time-series storage; high technical complexity and testing effort.

US-06: 8 pts — live streaming dashboard with WebSocket updates across many zones simultaneously; significant frontend and real-time-sync complexity.

US-07: 5 pts — builds on the ingestion pipeline with a new historical-query endpoint and charting UI; moderate effort.

US-08: 3 pts — rule-based heartbeat/fault checks on data already being ingested; well-understood pattern.

Total Sprint 1 Capacity Committed: 40 story points

This total falls within the team's historical velocity range of 38–42 points per two-week sprint (assumed for a 5-person cross-functional team), leaving a small buffer for code review, bug fixes, and sprint ceremonies.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

“Deliver a working foundation of the Intelligent Water Distribution Monitoring Platform where a utility admin can register, define DMAs/zones, invite team members, and view a live, real-time dashboard of flow and pressure data streamed from IoT sensors across the network — enabling an end-to-end demo of continuous network monitoring ahead of leak-detection analytics in Sprint 2.”

7.2 Expected Sprint Deliverables

- Fully functional utility registration, login, and organization setup module.
- DMA/zone creation flow with an interactive map-drawing tool and sensor-to-zone association.
- Team invitation system with role-based permissions (Operator, Field Technician, Manager, Analyst).
- Real-time sensor-data ingestion pipeline handling flow, pressure, and acoustic readings.
- Live monitoring dashboard showing zone-level status with color-coded alerts, refreshed continuously.
- Sensor drill-down view with historical trend charts for individual sensors.
- Automatic offline/faulty-sensor flagging to keep monitoring data trustworthy.
- A working end-to-end demo: live sensor data flowing from a simulated DMA into the real-time dashboard.

7.3 Definition of Done

- Code is peer-reviewed, merged to the main branch, and passes CI checks.
- Unit and integration tests written and passing for each story (minimum 80% coverage on new code).
- Acceptance criteria for each story verified by QA using the Given–When–Then scenarios in Section 3.
- Feature deployed to the staging environment and demonstrated at the Sprint Review.
- No critical or high-severity bugs open against the completed stories.

7.4 Sprint Timeline Overview

Day	Activity
Day 1	Sprint Planning — finalize sprint backlog and task assignments (this document).
Day 2–4	Utility onboarding, DMA/zone setup, and RBAC development.
Day 5–7	Sensor-ingestion pipeline and real-time dashboard implementation.
Day 8–9	Sensor drill-down view and offline/faulty-sensor flagging.
Day 10	Daily stand-ups throughout; mid-sprint backlog refinement.
Day 11–12	Integration testing, bug fixing, and QA verification against acceptance criteria.
Day 13	Sprint Review — demo to stakeholders.
Day 14	Sprint Retrospective — process improvement discussion.